

Machine Learning Basics

An introduction to ML concepts, algorithms, and model training with scikit-learn.

Welcome to the course! This document contains all the materials, lessons, and exercises you will need to successfully complete the curriculum. We recommend reading through this guide carefully and taking notes as you progress.

Category: Data | Level: Advanced | Est. Time: 15 Hours

Course Objectives & Overview

By the end of this course, you will have a solid understanding of the core concepts, practical applications, and best practices associated with this topic. This structured material is designed to guide you step-by-step from foundational theories to advanced problem-solving techniques.

Ensure you complete all practical exercises and review the lesson summaries before attempting the final exam.

Lesson 1: Introduction to Machine Learning

Machine learning enables computers to learn from data without explicit programming.

- Supervised vs. unsupervised vs. reinforcement learning
- Classification vs. regression problems
- Training set, validation set, test set
- Overfitting and underfitting
- Common algorithms: Linear Regression, Logistic Regression, Decision Trees, KNN, SVM
- Feature engineering and scaling
- Model evaluation: accuracy, precision, recall, F1, ROC-AUC
- scikit-learn pipeline: fit, transform, predict

Key concept: The goal of machine learning is to find patterns in data that generalize well to new, unseen data.

Key Takeaways:

- Review the main concepts outlined above.
- Practice the techniques discussed.
- Write down any questions for further research.

Conclusion & Next Steps

Congratulations on reaching the end of the written material for this course! Your next step is to log into the platform and take the final assessment. Review your personal notes, ensure you understand the key takeaways from each lesson, and apply what you have learned.

Good luck!